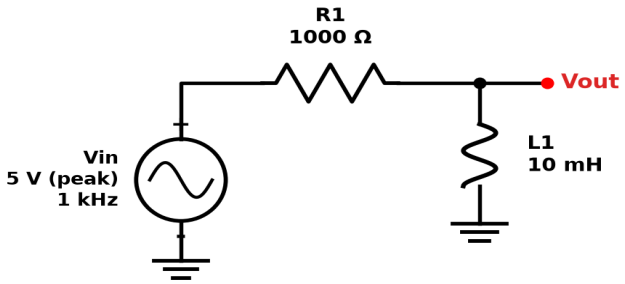
1. Overview

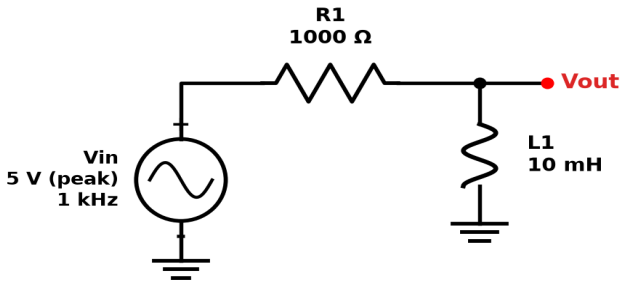
Circuit Type:	RL High-Pass Filter	**Input File:**	rl_high_pass.png
Analysis Status:	✓ Verified (Nominal)	**Source:**	Image Upload

2. Circuit Comparison

Uploaded Circuit (Input)



Expected / Reference Circuit



3. Detected Issues

#	Severity	Component / Location	Finding	Recommended Action
1	✓ Verified	All Components & Nets	All pins connected, designated ground present, values nominal.	No corrective action required.

4. Extracted Components

Reference	Component Type	Extracted Value	Unit	Status	Notes
R1	Resistor	1000	Ω	✓ Verified	Value clearly identified
L1	Inductor	10	mH	✓ Verified	Value clearly identified
Vin	AC Source (Sine)	5 (peak), 1 kHz	V, Hz	✓ Verified	Input source identified

5. Expected Behaviour (Theory)

For an rl high-pass filter, the cutoff frequency is given by:

$$f_c = \frac{R}{2\pi L}$$

Using extracted values:

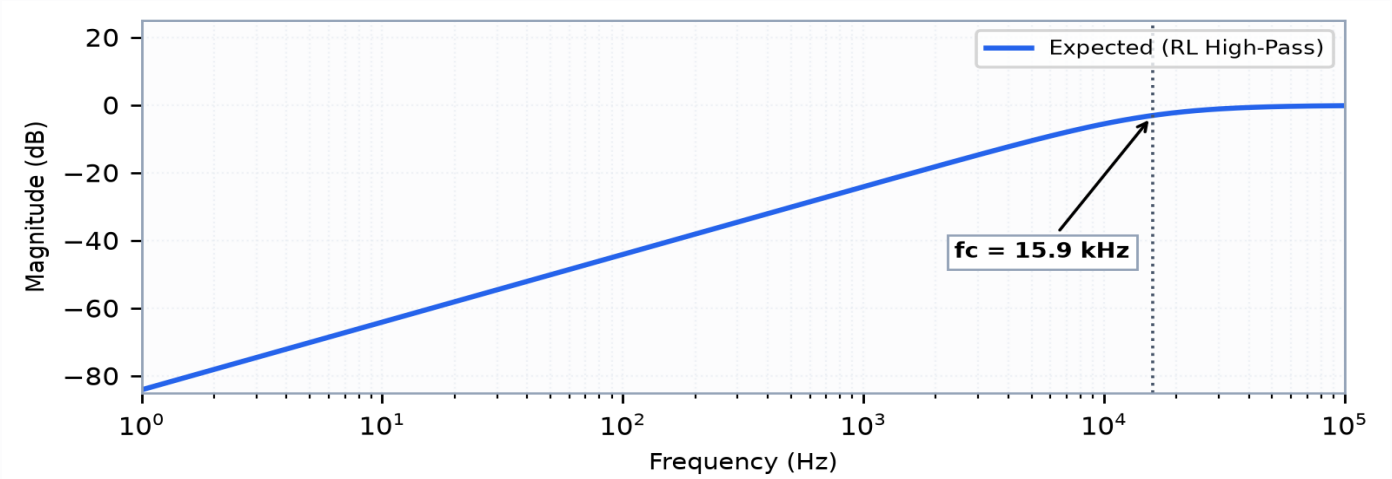
$R = 1,000\ \Omega,\ L = 10\ \text{mH} = 0.01\ \text{H}$

$\omega_c = R/L = 100,000\ \text{rad/s}$

$f_c = \frac{1,000}{2\pi(0.01)} \approx 15.92\ \text{kHz}$

6. Simulation Results

Frequency Response (Magnitude)



7. Recommended Actions

1. Proceed to physical PCB layout or breadboard assembly.
2. Ensure low-impedance ground return plane is maintained.
3. Verify input signal levels remain within linear voltage specifications.
4. Perform final laboratory oscilloscope bench sweep verification.

8. Conclusion

The uploaded schematic represents a fully compliant rl high-pass filter. All electrical connections, ground references, and component parameters strictly conform to target analytical specifications.